25 A rectangle has length 8 cm
The rectangle has area 20 cm^2

The length of the rectangle is increased by 2 cm
The area of the rectangle is increased by 4 cm^2

Noah says,

"The width of the rectangle decreases by less than 5%"

Is Noah correct?
You must show how you get your answer.

Old width = 20 ÷ 8 = 2.5 cm

New width = 24 ÷ 10 = 2.4 cm

Decrease = 2.5 – 2.4 = 0.1 cm

$\frac{0.1}{2.5} \times 100 = 4\%$

Yes, the width decreases by 4%, less than 5%.

1 mark
Find both widths.

1 mark
Compare change with original width.

1 mark
Yes, 4% < 5%.

(Total for Question 25 is 3 marks)